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REVIEW

Targeting the PI3K/AKT/mTOR pathway in triple-negative breast cancer: a review

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Abstract

Purpose Triple-negative breast cancer (TNBC) accounts for approximately 20% of breast cancer cases. Although there have been advances in the treatment of hormone receptor-positive and human epidermal growth factor receptor 2-positive breast cancers, targeted therapies for TNBC remain unavailable. In this narrative review, we summarize recent discoveries related to the underlying biology of the phosphoinositide 3-kinase (*PI3K*)/protein kinase B (*AKT*)/mechanistic target of rapamycin (*mTOR*) pathway in TNBC, examine clinical progress to date, and suggest rational future approaches for investigational therapies in TNBC.

Results As with other subtypes of breast cancer, aberrations in the *PI3K/AKT/mTOR* pathway are common in TNBC. Preclinical data support the notion that these aberrations predict TNBC inhibition by targeted agents. In a recently published phase 2 clinical trial, an AKT inhibitor (ipatasertib) improved outcomes in a subset of patients with metastatic TNBC when combined with paclitaxel in the first-line setting. In addition, new compounds with distinct specificity and potency targeting different PI3K/AKT/mTOR components and cognate molecules (e.g., mitogen-activated protein kinase) are being developed. These agents present a wide range of toxicity profiles and early efficacy signals, which must be considered prior to the advancement of new agents in later-phase clinical trials.

Conclusions The development of drugs targeting the PI3K/AKT/mTOR pathway for the treatment of TNBC is an evolving field that should take into account the efficacies and toxicities of new agents in addition to their interactions with different cancer pathways.

Keywords Triple-negative breast cancer · Targeted therapy · PI3K · mTOR

Purpose

Triple-negative breast cancer (TNBC) is defined by the absence of targetable aberrations, such as hormone receptor (HR) and human epidermal growth factor receptor 2 (HER2), represents up to 20% of all breast cancers, and is associated with poor clinical outcomes [1]. Despite multimodal treatments, patients with TNBC have a 3-year freedom-from-recurrence rate, which compares unfavorably with non-TNBC tumors (63 vs. 76%, $p < .0001$) [2, 3]. Chemotherapy continues to be the standard-of-care treatment for TNBC, with disappointing results; for example, chemotherapy regimens studied in the metastatic TNBC setting have shown overall response rates (ORRs) ranging from 10 to 35% and median progression-free survival (PFS) of only 3–4 months [4–6]. We performed a literature review to identify research that could have an impact on TNBC treatment. We describe recent discoveries related to the underlying biology of the phosphoinositide 3-kinase (*PI3K*)/protein kinase B (*AKT*)/mechanistic target of rapamycin (*mTOR*) pathway in TNBC, examine clinical progress to date, and suggest rational future approaches for investigational therapies in TNBC.

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Results

PI3K/AKT/mTOR pathway in TNBC

Aberrations in the *PI3K/AKT/mTOR* pathway are among the most common genomic abnormalities in breast cancer across various subtypes [7]. PI3Ks are pivotal molecules in this cascade that ultimately lead to tumor-cell growth (Fig. 1). PI3Ks are heterodimers composed of regulatory (p85) and catalytic (p110) subunits and exist in 4 isoforms (α, β, δ, and γ) [8]. The signaling pathway is activated by stimulation of receptor tyrosine kinases, which in turn trigger PI3K activation, followed by phosphorylation of AKT and mTOR complex 1 (mTORC1). In TNBC, oncogenic activation of the PI3K/AKT/mTOR pathway can happen as a function of overexpression of upstream regulators (e.g., epidermal growth factor receptor [EGFR]), activating mutations of *PI3K* catalytic subunit α (*PIK3CA*), loss of function or expression of phosphatase and tensin homolog (*PTEN*), and the proline-rich inositol polyphosphatase, which are downregulators of PI3K [9–11]. Conversely, activating mutations of downstream molecules to PI3K (i.e., *AKT* and *mTOR*) and cognate pathways (*MAPK* and *RAS*) are rare in TNBC [7]. Furthermore, several oncogenic pathways (i.e., FGFR, cMET, RAF) unleashed by *P53* inactivation converge to activate the PI3K pathway [12].

When compared to placebo, mTOR inhibitor everolimus in combination with aromatase inhibitor exemestane has improved the median PFS of patients with HR-positive (HR^+) HER2-negative ($HER2^-$) metastatic breast cancer (MBC, 10.6 vs. 4.1 months, $p < .001$) and is now a commonly used treatment option [13]. PI3K/AKT/mTOR-targeted therapies have not yet been successfully developed for TNBC, although an early-phase clinical trial has suggested the efficacy of agents targeting this pathway. Ipatasertib (an AKT inhibitor) combined with paclitaxel improved the PFS of patients with TNBC when compared to paclitaxel alone (6.2 vs. 4.9 months, $p = .037$), indicating that PI3K pathway-targeted therapies may lead to promising results in a subset of patients [14].

Genomic aberrations in TNBC

TNBC samples can be grouped as basal-like breast tumors, which present gene expression cluster characteristics of basal epidermal cells, including keratin 5, keratin 17, integrin-β4, and laminin [15]. In addition, the mutational landscape of TNBC was analyzed through whole-exome sequencing [7]. Basal-like breast cancers showed staggering genomic heterogeneity epitomized by low mutational

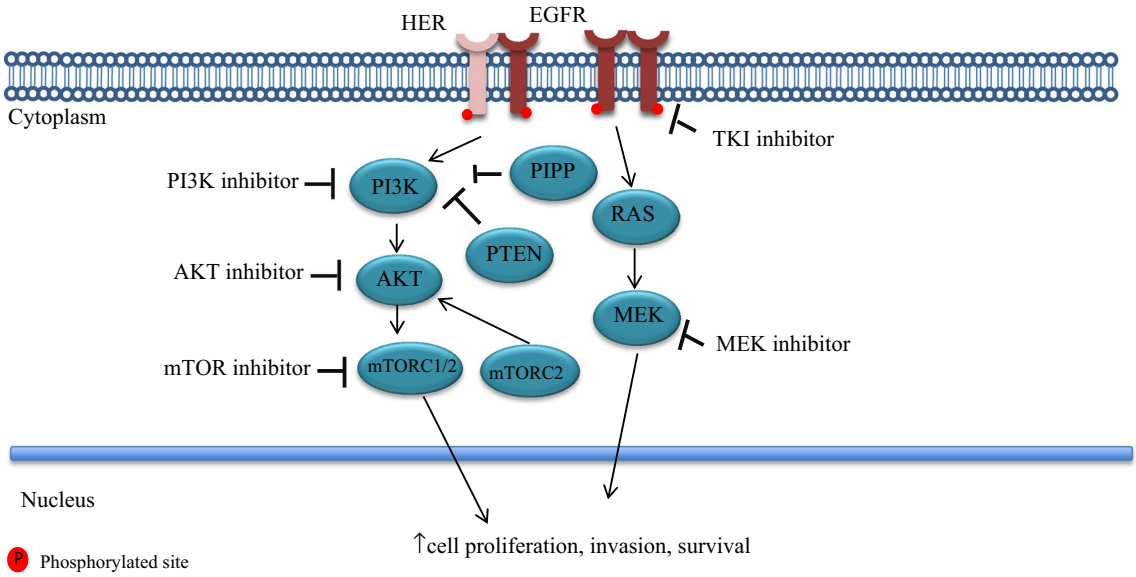

Fig. 1 Potential targets in the PI3K/AKT/mTOR pathway for drug development for triple-negative breast cancer. PI3K/AKT/mTOR signal transduction: cytoplasmatic activation of PI3K is secondary to activating mutations in the α catalytic unit and/or *PTEN* and/or *PIPP* loss. Alternatively, activation of the receptor tyrosine kinase occurs by homodimerization or heterodimerization with other HER family members, such as HER2. Activated EGFR initiates downstream signaling through the PI3K/AKT/mTOR and RAS-MEK pathways, promoting cell proliferation and survival. Lines with flat ends represent the inhibitory activity of targeted therapies, and lines with arrowheads represent the stimulatory activity of upstream molecules. *ADC* Antibody drug conjugate, *EGFR* epidermal growth factor receptor, *HER* human epidermal receptor, *mTORC* mechanistic target of rapamycin complex, *PTEN* phosphatase and tensin homolog, *PIPP* prolin-rich inositol polyphosphatase, *TKI* tyrosine kinase inhibitor

frequencies (i.e., < 5%) in all genes except tumor suppressor genes *P53* (80% of basal-like tumors) and *PIK3CA* (9%).

In a seminal work, Lehmann et al. showed that TNBC could be clustered based on gene ontologies, with the expression profiling of 587 tumor samples (386 in the training set and 201 in the validation set) [16]. Despite significant heterogeneity among TNBC samples, 6 TNBC types were proposed: basal-like 1, basal-like 2, immune modulatory, mesenchymal, mesenchymal stem-like (MSL), and luminal androgen receptor. The basal-like 2 subtype displayed cognate genes involving growth factors (epidermal growth factor, MET, Wnt/β-catenin, and insulin growth factor receptor 1) and growth factor receptors (EGFR, MET receptor, and EPAH2). These receptors and cognate downstream molecules have been implicated in the PI3K/AKT/mTOR-signaling pathway. Most importantly, TNBC cell lines classified as luminal androgen receptor, and MSL were found to frequently harbor *PIK3CA* and *PTEN* mutations and exhibited significant sensitivity to PI3K and mTOR inhibition by NVP-BEZ235 (a dual PI3K and mTOR inhibitor) both in vivo and in vitro, indicating that the PI3K/AKT/mTOR pathway drives the biology of different subtypes of TNBC. Smaller studies also support the presence of PI3K/AKT/mTOR aberrations in a subset of patients with TNBC, as 40–70% of cases present overexpression of phosphorylated mTOR protein and approximately 22% present with *PIK3CA* mutations [17–19]. Furthermore, as many as 48% of patients with metastatic TNBC have tumors harboring low PTEN expression [14].

Targeting PI3K/AKT/mTOR in TNBC

PI3K inhibitors in TNBC

PI3K inhibitors exhibit different potencies against distinct PI3K types (i.e., α, β, δ, and γ subunits). PIK3CA is the most commonly activated isoform in breast cancer, including TNBC. Thus far, class IA pan-PI3K inhibitors have been more extensively studied than isoform-specific inhibitors. For example, the clinical efficacy of buparlisib (BKM120), a PI3K antagonist with activity against all PI3K isoforms, has already been assessed in a phase 3 trial [20]. In the BELLE-2 trial (Baselga et al.), a total of 1147 patients with progressive HR^+ $HER2^-$ MBC after treatment with antiestrogen therapy were randomized to treatment with fulvestrant at standard dosage, with or without buparlisib (100 mg/day). The median PFS was 6.9 months in the buparlisib group versus 5 months in the placebo group ($p = .00021$) [21]. Interestingly, among the 372 patients with tumors harboring PI3K activation (i.e., mutations on *PIK3CA* exons 1, 7, 9, or 20 and/or immunohistochemistry-based PTEN loss), buparlisib improved the median PFS from 4 to 6.8 months ($p = .014$). The most common adverse events (AEs) in the buparlisib cohort were grade ≥ 3 alanine aminotransferase (25%) and aspartate aminotransferase elevations (18%), hyperglycemia (15%), and rash (8%). Buparlisib is currently under investigation for the treatment of TNBC in a phase 2 trial (NCT01790932).

Pictilisib is another pan-PI3K inhibitor with greater subunit α-inhibitor activity than buparlisib (IC50 = 3 nM versus IC50 = 52 nM, respectively), which showed a favorable toxicity profile in a multiple histology, phase 1 trial. The most common AEs were grades 1–2 nausea, rash, and fatigue [22, 23]. In a randomized phase 2 trial, however, pictilisib was given at 2 different dose levels (340 mg and 260 mg) in combination with fulvestrant for the treatment of progressive HR^+ $HER2^-$ MBC after antiestrogen therapy [24]. At the 340 mg dose, grade ≥ 3 AEs occurred in 54 out of 89 (61%) patients, and at the 260 mg dose level, grade ≥ 3 AEs occurred in 15 out of 42 (36%) patients. Diarrhea and skin rash were the most common AEs in both groups. These results suggest that agents with greater PI3K specificity may lead to higher frequencies of AEs when compared to less potent agents. As preclinical studies continue to explore the efficacy of pictilisib alone or in combination with other targeted agents (such as the insulin growth factor receptor 1 pathway) in TNBC models, translational trials will need to account for the higher absolute risk of AEs associated with this compound [25]. Finally, although other pan-PI3K inhibitors (e.g., SAR245408, taselisib, and PX-866) are under development for the treatment of $HER2^+$ and HR^+ breast cancer subtypes, studies in the realm of TNBC remain lacking [26–28].

Isoform-specific PI3K inhibitors are being tested in early-phase clinical trials. Alpelisib is an oral class I α-specific PI3K inhibitor compound, which has shown preliminary evidence of antitumor activity for the treatment of HR^+ $HER2^-$ MBC [29]. In a phase 1b trial, 26 patients with antiestrogen refractory MBC were treated with letrozole and alpelisib [30]. At the maximum-tolerated dose of 300 mg daily, 2 patients experienced grade 3 AEs of diarrhea, 2 of hyperglycemia, and 1 of transaminase elevation. Among patients with *PIK3CA*-mutated tumors, the 6-month clinical benefit rate was 44%, while it was 20% among patients with *PIK3CA* wild-type tumors. Grade 3 toxicities associated with alpelisib were uncommon, and transaminitis and gastrointestinal side effects were less frequent than for pan-PI3K inhibitors. As alpelisib continues to be developed for the treatment of HR^+ $HER2^-$ and $HER2^+$ MBC (NCT03056755, NCT02167854), preclinical data support the notion that alpelisib induces decreased AKT phosphorylation across different TNBC subtypes (i.e., basal-like 1, mesenchymal, and MSL) [31]. Furthermore, a phase 1b trial of enzalutamide combined with alpelisib for the treatment of TNBC is ongoing (NCT03207529).

Serabelisib is a potent and selective PIK3CA inhibitor (IC50 21 nM) that is under development for the treatment of solid tumors [32]. In a phase 1 trial, a total of 71 patients received at least 1 dose of serabelisib, and 54% of patients reported at least 1 ≥ grade 3 AE, indicating that intermittent dosing was required [32]. Sixteen patients had MBC (type not reported), 3 of whom achieved a partial response; all 3 had tumors harboring *PIK3CA* mutations.

AKT inhibitors in TNBC

AKT is downstream to PI3K (Fig. 1) and is inhibited by ipatasertib [33]. In the recently published LOTUS (LOng-Term follow-Up Study) trial, a total of 124 patients with untreated metastatic TNBC were randomized to 80 mg/m^2 paclitaxel (on days 1, 8, and 15) with 400 mg oral ipatasertib or placebo (days 1 through 21 on 28-day cycles) [14]. The median PFS was significantly improved among molecularly unselected patients treated with ipatasertib compared to placebo (6.2 vs. 4.9 months, hazard ratio 0.6, $p = .037$). Grades 3–4 diarrhea (23%), neutropenia (10%), and stomatitis (2%) were observed. No treatment-related deaths, pneumonitis, or significant weight loss were reported, which compares favorably with the performance of the FDA-approved mTOR inhibitor everolimus [13]. In addition, the absolute risk of grade ≥ 3 AEs was only modestly greater among patients treated with ipatasertib than among those treated with paclitaxel alone (54 vs. 42%, respectively), and diarrhea was the most common grade 3–4 AE in the ipatasertib arm (23%). PFS showed significant correlation with *PI3K/AKT/PTEN* alterations (hazard ratio 0.44, $p = .041$) indicating that the *PI3K/AKT/mTOR* pathway indeed drives the biology of a subset of patients with TNBC. Other AKT inhibitors are already under clinical development as monotherapy or in combination with chemotherapy or targeted agents for the treatment of MBC, including TNBC (e.g., AZD5363, MK-2206) [34–36].

Mechanistic target of rapamycin inhibitors in TNBC

Data from 177 patients with TNBC showed that 72% of tumors harbored phosphorylated mTOR [19]. After of a median follow-up period of 69 months, univariate analysis showed significant correlation of phosphorylated mTOR with shorter overall- and recurrence-free rates of survival for patients with stages I and II TNBC ($p = .001$ and .004, respectively). A patient-derived xerograph TNBC model testing the mTOR inhibitor rapamycin showed 77–99% tumor-growth inhibition, which is significantly more than that has been seen with doxorubicin; protein phosphorylation studies indicated that constitutive activation of the mTOR pathway decreased with treatment [37].

More recently, 52 women with metaplastic TNBC were treated with liposomal doxorubicin, bevacizumab, and temsirolimus or everolimus in a phase 1 trial. Among unselected patients, the ORR was 21% and the clinical benefit rate at 6 months was 40%. Tissue samples from 43 patients were available for testing, and 32 (74%) were found to have a PI3K pathway aberration. The presence of activating PI3K pathway aberrations was associated with a significant improvement in ORR (31 vs. 0%, $p = .04$) but not the clinical benefit rate [38]. As most patients enrolled in this trial had received prior anthracycline-based (75%) and/or taxane-based therapies (81%), the results compare favorably with historical controls.

Targeting both mTORC1 and mTORC2 has the theoretical advantage of blocking the feedback loop leading to AKT activation by mTORC2 upon inhibition of mTORC1 [39]. Preclinical data support in vivo activity and induction of decreased levels of p-AKT by mTORC1/2 inhibitor AZD2014, when compared with everolimus in TNBC patient-derived xerograph models [39]. Other mTORC1/2 inhibitors have shown antitumor activity in vitro across different breast cancer cell lines, including TNBC (e.g., MLN0128) [40].

Dual PI3K/AKT/mTOR inhibitors

As the blockade of mTOR gives rise to a mechanism of negative feedback that leads to the activation of upstream molecules, it is hypothetically possible that blockade of the PI3K/AKT/mTOR pathway can lead to increased antitumor activity in TNBC [41]. Preclinical data have shown that the combination of compounds targeting different cognate molecules in the PI3K/AKT/mTOR pathway leads to synergistic activity. Xu et al. reported results of a *PTEN*-knocked basal-like breast cancer model in which the combination of MK-8669 (mTOR inhibitor) and MK-2206 (AKT inhibitor) improved the mean percentage of tumor-growth inhibition compared to either one alone in vivo ($p < .001$) [42].

On the basis of these findings, new compounds targeting different components of the PI3K/AKT/mTOR pathway simultaneously continue to be developed. For example, gedatolisib inhibits mutant forms of PI3K-α with elevated kinase activity at concentrations equivalent to the IC50 for wild-type PI3K-α. PI3K-β, -δ, and -γ isoforms were inhibited by gedatolisib at concentrations approximately 10-fold higher than those observed for PI3K-α. Mechanistic target of rapamycin kinase inhibition (IC50 = 1 nmol/L) indicated that gedatolisib was an equipotent PI3Kα/mTOR inhibitor and is more potent than currently approved mTOR inhibitor everolimus [43]. Another advantage of simultaneously targeting PI3K and mTOR is the ensuing more robust inhibition of receptor tyrosine kinase-positive feedback loops seen with isolated PI3K inhibition [44]. TNBC cell line

MDA-MB-231 showed significant inhibition by gedatolisib in vitro (IC50 0.042 μmol/L). The safety and antitumor activity of gedatolisib have been assessed in early-phase clinical trials among patients with endometrial cancer, and the drug appears to have a favorable toxicity profile [45, 46]. Gedatolisib is currently under development for the treatment of TNBC, in combination with PTK7 antibody–drug conjugate (NCT03243331).

Apitolisib (GDC-0980) is a PI3K inhibitor (subunits α, δ, and γ) that also targets mTORC [47]. In a cell-derived TNBC model (MDA-MB-231), mice treated with GDC-0980 achieved tumor-growth inhibition of over 50% [48]. A phase 1 study is ongoing (NCT00854126). Omipalisib is a potent pan-PI3K and mTORC inhibitor that showed in vivo activity in a $HER2^+$ model (ABT414 cells) [49].

Preclinical data continue to propel other dual PI3K/AKT/mTOR inhibitors into early-phase clinical trials, but few have moved specifically to clinical trials for TNBC. For instance, omipalisib (GSK2126458) is still in its initial stages of clinical development in a phase 1 multiple-histology trial (NCT00972686). Voxtalisib (SAR245409, XL765) is a PI3K (subunits α, β, δ, and γ) and mTOR inhibitor with clinical trials underway for its use in the treatment of HR^+ tumors only [50]. Other PI3K inhibitors with high affinities to isoforms that are not found in TNBC (e.g., isoform δ) have been developed for the treatment of hematologic malignancies [51].

Targeting PI3K/AKT/mTOR pathway interactions in TNBC

Epidermal Growth Factor Receptor

EGFR is upstream to PI3K and activates this pathway [52]. Results of in vitro studies of MDA-MB-231 and MDA-MB-468 TNBC cell models treated with rapamycin (0.78–100 nmol/L) and/or lapatinib (0.156–20 nmol/L) showed that in both cell lines, lapatinib alone produced a gradual dose-dependent growth inhibition, and the lapatinib-rapamycin combination produced significant dose-dependent cell growth inhibition. In CDX models (MDA-MB-468 and MDA-MB-231) the combination of lapatinib and rapamycin also showed synergistic antitumor effects [9].

Mitogen-activated protein kinase kinase

Analysis of TNBC tumor specimens showed high levels of activation of the RAS/MEK/extracellular signal-regulated kinase (ERK) pathway, supporting MEK as a suitable target for therapeutic intervention in a subset of TNBC patients [53]. Mirzoeva et al. demonstrated that in the presence of epidermal growth factor, MEK inhibitor-induced activation of AKT (by feedback activation of PI3K induced by MEK blockade) is abrogated by the treatment of MDA-MB-231 cells with gefitinib, indicating cross talk between PI3K and MEK pathways [54]. Dactolisib (BEZ235) is a PI3K (subunits α, β, δ, and γ) and mTOR inhibitor that is currently under development in combination with MEK162 (MEK inhibitor) for the treatment of TNBC (NCT01337765) [55].

Poly (ADP-ribose) polymerase

Deregulation of BRCA1, a protein with critical roles in the homologous recombination-dependent DNA repair pathway, has been attributed to several mechanisms, including BRCA1-promoter methylation and overexpression of the negative regulators ID4 and HMG [56–58]. The premise for the synthetic lethality approach is tumor sensitivity to blockade of DNA single-strand breaks by poly (ADP-ribose) polymerase (PARP) inhibition [59]. Under normal conditions, the PI3K/AKT/mTOR pathway stabilizes and preserves DNA double-strand break repair by interacting with the homologous recombination complex, and it is necessary for DNA repair during ionizing radiation [60]. In a TNBC cell model, PI3K inhibition with buparlisib impairs homologous recombinant DNA repair, inducing further genomic aberrations [61]. De et al. showed that the combination of GDC-0980 (pan-PI3K and mTOR inhibitor) with ABT888 (inhibitor of PARP) led to significant tumor-growth inhibition in a *PTEN*-null in vivo TNBC model (MDA-MB468) [62]. Mechanistic target of rapamycin complex 1/2 inhibitor AZD2014 in combination with PARP inhibitor olaparib is currently under the initial stages of clinical development for the treatment of TNBC (NCT02208375).

PI3K/AKT/mTOR and immunotherapy in TNBC

Immune checkpoint inhibitor antibodies selectively block the interaction between programed cell death 1 (PD-1) and PD-1 ligand 1 (PD-L1) that are expressed on cytotoxic $CD8^+$ T cells and tumor cells, respectively, leading to the activation of cellular immunity. Pembrolizumab, a monoclonal anti-PD-1 antibody, was tested in a phase 1b trial involving 32 heavily pretreated women with PD-L1 IHC^+ recurrent metastatic TNBC; the ORR was 18.5% [63]. Avelumab, a PDL-1 inhibitor, also showed signs of preliminary activity in patients with MBC in a phase 1b trial [64]. Forty-four percent of patients (4 of 9) who had $PD\text{-}L1^+$ immune cells within the tumor had partial responses, whereas 2.6% of TNBC patients (1 of 39) with $PD\text{-}L1^-$ immune cells had the same outcome. The mutational burden of tumors correlates with clinical benefit from immune checkpoint inhibitors, with the tumors displaying the highest rates of mutations having remarkable antitumor effects [65]. In TNBC cells, PI3K inhibition impairs homologous recombinant DNA repair, inducing further genomic aberrations, which

could lead to increases in sensitivity to immune checkpoint antibodies [61]. Evidence also supports regulation of the adaptive cytotoxic immune system through upregulation of PD-L1 via activation of PI3K/AKT/mTOR in TNBC. Mittendorf et al. showed that *PTEN* loss (induced by PTEN antiviral PTEN small hairpin RNA vector) led to increased PD-L1 cell surface expression in a MDA-MB-157 cell line ($p < .001$) [66]. *PTEN* knockdown in the MDA-MB-231 cell line resulted in a greater increase in PD-L1 expression than the addition of IFN-γ, which is known to enhance PD-L1 expression. Most importantly, treatment of those TNBC cells with AKT inhibitor or mTOR (MK2206 or rapamycin, respectively) reduced expression of PD-L1 ($p < .01$). An early-phase clinical trial is currently assessing the safety and efficacy of an immune checkpoint inhibitor (PDR001) combined with an mTOR inhibitor (everolimus) in patients with TNBC (NCT02890069).

Conclusions

The PI3K/AKT/mTOR pathway drives the biology of a subset of TNBC, with the obvious caveat that this represents a minority of cases [16]. Preclinical and multiple-histology early-phase clinical trials suggest that careful selection of compounds based on their specificity and potency to signaling pathways of interest is mandatory when designing early-phase clinical trials assessing the safety and efficacy of new agents. This is illustrated by the example of a phase 2 trial comparing gedatolisib (pan-PI3K and mTORC inhibitor) and PF-04691502 (pan-PI3K and AKT inhibitor) for the treatment of recurrent endometrial tumors [46]. Enrollment was stopped due to unacceptable toxicities that included pneumonitis and pneumonia and the deaths of 9 study participants. The 9 deaths occurred during the course of the study while patients were receiving PF-04691502, and thus far, gedatolisib has demonstrated favorable toxicity.

Results of the LOTUS trial are worth careful consideration. Notwithstanding the modest absolute improvement in clinical outcomes seen in patients treated with ipatasertib (median PFS of 6.2 months), important lessons were learned

Table 1 Ongoing clinical trials assessing safety and efficacy of PI3K/AKT/mTOR-targeted agents in TNBC

Compound	Additional agent	Compound inhibitory selectivity						Study phase	Trial ID
		PI3K-α	PI3K-β	PI3K-δ	PI3K-γ	AKT	mTOR		
BYL719 [29]	NA	√						2	NCT02506556
Taselisib [67]	Enzalutamide (non-steroidal antiandrogen)	√	√	√	√			1b/2	NCT02457910
Gedatolisib [43]	Paclitaxel, docetaxel, dacomitinib (pan-HER inhibitor)	√					√	1b	NCT01920061
AZD8186 [68]	Docetaxel	√	√	√				1	NCT03218826
AZD8186 [68]	AZD2014 (AKT and mTOR inhibitor)	√	√	√				1	NCT01884285
Gedatolisib [43]	PTK7-ADC	√			√			1	NCT03243331
Everolimus [69]	PDR001 (PD-1 monoclonal antibody)						√	1	NCT02890069
Everolimus [69]	Vinorelbine			√			√	2	NCT01520103
Everolimus [69]	Erlotinib (EGFR inhibitor)			√				1/2	NCT00574366
Everolimus [69]	Eribulin			√				1	NCT02616848
AZD2014 [39]	Selumetinib (MEK inhibitor)					√	√	1/2	NCT02583542
SAR245409 [50]	MSC1936369B (MEK inhibitor)	√	√	√	√			1b	NCT01390818
MK2206 [70]	NA					√		2	NCT01277757
AZD5363 [71]	Paclitaxel					√		2	NCT02423603
GSK2141795 [72]	Trametinib (MEK inhibitor)					√			NCT01964924
Ipatasertib [33]	Paclitaxel							2	NCT02301988
AZD2014 [39]	Olaparib (PARP inhibitor)					√	√	1/2	NCT02208375
Temsirolimus [73]	Neratinib (EGFR and HER2 inhibitor)						√	1/2	NCT01111825

Data extracted from www.clinicaltrials.gov, accessed on November 2, 2017. Includes multiple histology, all breast cancers, and exclusively TNBC trials

ADC antibody drug conjugate, *AKT* protein kinase B, *EGFR* epidermal growth factor receptor, *HER* human epidermal growth factor receptor, *ID* identification, *mTOR* mechanistic target of rapamycin, *NA* not applicable, *PARP* poly (ADP-ribose) polymerase, *PD-1* programed cell death 1, *PI3K* phosphoinositide 3-kinase, *PTK7* protein-tyrosine kinase-like 7, *TNBC* triple-negative breast cancer

from this study: (i) the magnitude of benefit was greater among patients with *PI3K/AKT/mTOR*-aberrant tumors, indicating the need for more personalized strategies and (ii) 48% of patients had low expression of PTEN that did not overlap with genomic aberrations for many, suggesting that proteomic aberrations should also be explored as predictive markers in future trials.

Finally, moving forward, the field of developmental therapeutics needs to take into account preclinical evidence of significant interactions between PI3K/AKT/mTOR and other oncogenic signaling pathways and cognate molecules, such as transmembrane receptors and DNA repair pathways (Table 1). Also, a subset of TNBCs has prominent expression of immune-modulatory genes; indeed, early-phase trials already support activity of immune checkpoint inhibitors in a small subset of patients with TNBC [16, 63, 64]. Data on the intersection of the PI3K/AKT/mTOR-signaling pathway and the adaptive immune system in TNBC are beginning to emerge.

In conclusion, the development of drugs targeting the PI3K/AKT/mTOR pathway for the treatment of TNBC is an evolving field that should take into account the efficacies and toxicities of new agents in addition to their interactions with different cancer pathways.

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Compliance with ethical standards

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